

Experiment No. 4

WAVELENGTH OF LASER USING DIFFRACTION GRATING

Aim: To determine the wavelength of a given laser source using diffraction grating.

Apparatus: Diffraction Grating, laser source, screen, scale and stand.

Formula-Diffraction condition for maxima is,

$$n\lambda = d \sin\theta$$

Where,

d = grating constant (m)

θ = Angle of diffraction (degree)

n = order of Spectrum

λ = wavelength of laser source (m)

Grating Constant:

The distance between two adjacent slits of a diffraction grating is known as grating constant.

Its value is obtained by dividing the length of grating by the total number of lines ruled on the grating or if number of lines per unit length is given then it will be reverse of that in that unit.

$d = L / N$ (m)

L = length of the grating (m)

N = number of lines ruled on the grating

Observation:

No. of lines per inch on grating is (N) = 500,

1 inch = 2.54×10^{-2} m

$$d = \frac{2.54 \times 10^{-2}}{500} \text{ m}$$

Theory:

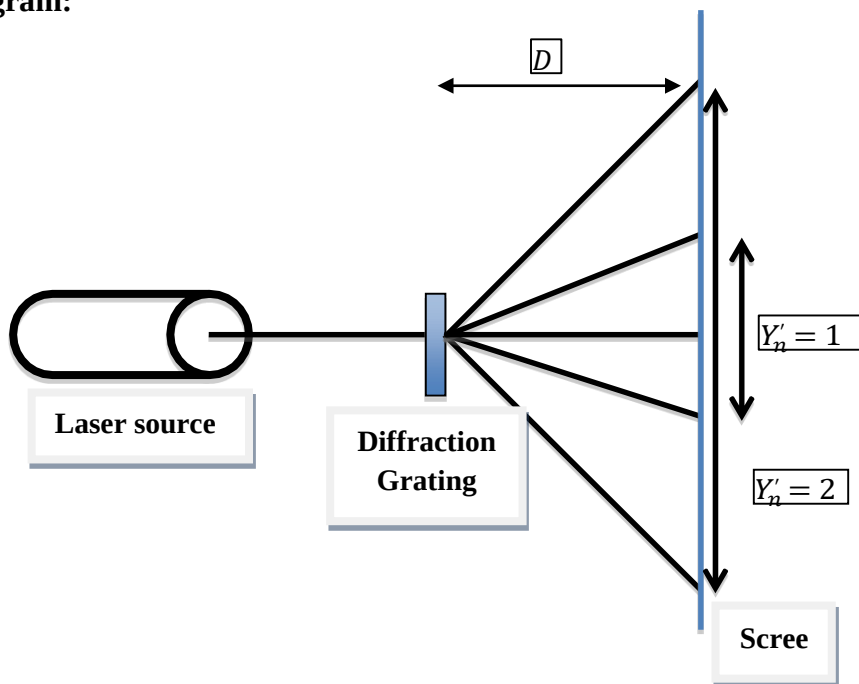
A diffraction grating consists of a large number of parallel slits of equal width separated from one another by equal opaque spacing. When monochromatic beam of light falls on this grating it splits it in to different

beams. These beams superimpose in the medium resulting a diffraction pattern on the screen placed in front of grating. By measuring the position of maxima using above relation one can determine the wavelength of the light used.

Procedure:

1. Mount the grating on the grating holder and set it parallel to the screen.
2. The grating should be perfectly vertical and set for normal incidence.
3. Measure distance Y_m of n^{th} order spot on the screen.
4. Measure distance between the grating and the screen; denote it as (D).
5. Repeat the procedure for another value of distance (D) between the grating and the screen.

Diagram:



OBSERVATION TABLE:

Sr. No.	Order of Spectrum n	D cm	Distance between the points of same order on either side of central maxima. Y' (cm)	Y=Y'/2 (cm)	$\sin\theta = \frac{Y}{(D^2 + Y'^2)^{1/2}}$	Wavelength λ (nm)*
1	1					
2	2					
3	1					
4	2					
5	1					
6	2					

$$1 \text{ nm} = 10^{-9} \text{ m or } 10^{-9} \text{ m} = 1 \text{ nm}$$

Calculation:**Result:**

Wavelength of the given laser source is _____ nm.

Precaution:

1. Keep the distance between grating and source as constant.
2. Don't expose the laser light to the eyes.
3. Laser light should fall on the screen normally.